

STA 445 S26 Assignment 2

2026-02-24

Load all the packages you will need here. Notice that the code chunk option has been set to include = FALSE. This means the code will run, but the output will not be shown in the knitted document. Thus, you do not need to worry about suppressing warning messages. This code will be helpful when writing a project report.

Problem 1

Download from GitHub the data file by clicking on this link: [Example_5.xls](#). Open it in Excel and figure out which sheet of data we should import into R. At the same time figure out which range of values to import.

- a. Import the dataset into a data frame and show the structure of the imported data using the `str()` command. Make sure that your data has $n = 31$ observations and the three columns are appropriately named. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file.

```
df1 <- read_excel("Example_5.xls", "RawData", "A5:C36")
str(df1)
```

```
## tibble [31 x 3] (S3: tbl_df/tbl/data.frame)
##  $ Girth : num [1:31] 8.3 8.6 8.8 10.5 10.7 10.8 11 11 11.1 11.2 ...
##  $ Height: num [1:31] 70 65 63 72 81 83 66 75 80 75 ...
##  $ Volume: num [1:31] 10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 ...
```

- b. Use `slice_sample()` to show five random rows of the dataset.

```
slice_sample(df1, n=5)
```

```
## # A tibble: 5 x 3
##   Girth Height Volume
##   <dbl> <dbl> <dbl>
## 1  14.2     80   31.7
## 2   8.6     65   10.3
## 3  12       75   19.1
## 4   8.3     70   10.3
## 5  11       75   18.2
```

Problem 2

Download from GitHub the data file by clicking on this link: [Example_3.xls](#). Import the data set into a data frame and show the structure of the imported data using the `str()` command. Make sure the Tesla values are NA where appropriate and that both -9999 and NA are imported as NA values. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file.

```
df2 <- read_excel("Example_3.xls", "data", "A1:L34", na=c("NA", -9999))
str(df2)
```

```
## tibble [33 x 12] (S3: tbl_df/tbl/data.frame)
## $ model: chr [1:33] "Mazda RX4" "Mazda RX4 Wag" "Datsun 710" "Hornet 4 Drive" ...
## $ mpg : num [1:33] 21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
## $ cyl : num [1:33] 6 6 4 6 8 6 8 4 4 6 ...
## $ disp : num [1:33] 160 160 108 258 360 ...
## $ hp : num [1:33] 110 110 93 110 175 105 245 62 95 123 ...
## $ drat : num [1:33] 3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
## $ wt : num [1:33] 2.62 2.88 2.32 3.21 3.44 ...
## $ qsec : num [1:33] 16.5 17 18.6 19.4 17 ...
## $ vs : num [1:33] 0 0 1 1 0 1 0 1 1 1 ...
## $ am : num [1:33] 1 1 1 0 0 0 0 0 0 0 ...
## $ gear : num [1:33] 4 4 4 3 3 3 3 4 4 4 ...
## $ carb : num [1:33] 4 4 1 1 2 1 4 2 2 4 ...
```

b. Use the `tail()` command to show the last five rows of a data table.

```
tail(df2)
```

```
## # A tibble: 6 x 12
##   model      mpg   cyl  disp    hp  drat    wt   qsec    vs  am gear carb
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Lotus Europa 30.4     4  95.1   113  3.77  1.51  16.9     1   1   5    2
## 2 Ford Panter~ 15.8     8  351    264  4.22  3.17  14.5     0   1   5    4
## 3 Ferrari Dino 19.7     6  145    175  3.62  2.77  15.5     0   1   5    6
## 4 Maserati Bo~ 15       8  301    335  3.54  3.57  14.6     0   1   5    8
## 5 Volvo 142E   21.4     4  121    109  4.11  2.78  18.6     1   1   4    2
## 6 Tesla Model~ 98      NA   NA    778 NA    4.94  10.4    NA   0   1   NA
```

c. Use `slice_sample()` to show five random rows of the dataset.

```
slice_sample(df2, n=5)
```

```
## # A tibble: 5 x 12
##   model      mpg   cyl  disp    hp  drat    wt   qsec    vs  am gear carb
##   <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Porsche 914~ 26       4  120.    91  4.43  2.14  16.7     0   1   5    2
## 2 Ford Panter~ 15.8     8  351    264  4.22  3.17  14.5     0   1   5    4
## 3 Chrysler Im~ 14.7     8  440    230  3.23  5.34  17.4     0   0   3    4
## 4 Pontiac Fir~ 19.2     8  400    175  3.08  3.84  17.0     0   0   3    2
## 5 Valiant     18.1     6  225    105  2.76  3.46  20.2     1   0   3    1
```

Problem 3

Download all of the files from GitHub `data-raw/InsectSurveys` directory which have been linked here. Each month's file contains a sheet with information about each of the sites that was surveyed. The second sheet contains information about the number of each species that was observed at each site.

- a. Import the data for each month and create a single `site` data frame with information from each month. You are not allowed to make any modifications to the data file. All changes must be made within this rmd file. You may need to change column names and reorganize columns.

```
sep <- read_excel("september.xlsx", "sites", "A1:F10", .name_repair=toupper)
aug <- read_excel("august.xlsx", "sites", "A1:F10", .name_repair=toupper)
jul <- read_excel("july.xlsx", "Sites", "A1:F10", .name_repair=toupper)
jun <- read_excel("june.xlsx", "Sites", "A1:F10", .name_repair=toupper)
may <- read_excel("may.xlsx", "Sites", "A1:F10", .name_repair=toupper)
oct <- read_excel("october.xlsx", "Sites", "A1:F10", na="Did not vist", .name_repair=toupper)
site <- rbind(sep, aug, jul, jun, may, oct)
```

- b. Show the structure of the data frame using the `str()` command.

```
str(site)
```

```
## tibble [54 x 6] (S3: tbl_df/tbl/data.frame)
##  $ SITE NAME   : chr [1:54] "Araphahoe Road" "Bridger Valley" "Calculus Vector" "Deer Valley" .
##  $ POND AREA   : num [1:54] 34 240 321 74 28 62 489 398 126 34 ...
##  $ WATER DEPTH: num [1:54] 3 6 13 4.4 2 3.6 4 10 9 3 ...
##  $ PH          : num [1:54] 6.2 6.5 6.4 6.9 7.1 7 7.1 6.8 6.75 6.2 ...
##  $ DATE        : POSIXct[1:54], format: "2020-09-15" "2020-09-16" ...
##  $ OBSERVER    : chr [1:54] "Bob" "Bob" "Bob" "Bob" ...
```

- c. Use `slice_sample()` to show eight random rows of the dataset.

```
slice_sample(site, n=8)
```

```
## # A tibble: 8 x 6
##   'SITE NAME'      'POND AREA' 'WATER DEPTH'    PH DATE              OBSERVER
##   <chr>            <dbl>        <dbl> <dbl> <dtm>              <chr>
## 1 Happy Feet      398          10    6.8 2020-07-18 00:00:00 Charlie
## 2 Gigantic Pain  489           4    7.1 2020-07-17 00:00:00 Charlie
## 3 Ephemeral Stream 28            2    7.1 2020-06-15 00:00:00 Charlie
## 4 Ephemeral Stream 28            2    7.1 2020-09-15 00:00:00 Charlie
## 5 Fennel Gardens  62           3.6    7    NA              Charlie
## 6 Happy Feet      398          10    6.8 2020-09-18 00:00:00 Charlie
## 7 Gigantic Pain  489           4    7.1 2020-08-17 00:00:00 Charlie
## 8 Fennel Gardens  62           3.6    7    2020-05-16 00:00:00 Charlie
```

- d. Comment on the importance of consistency of your data input sheets.

Inconsistent data input will cause significant difficulty in getting your data into a usable format. In some cases such as this one this will mean errors and broken code, but in other cases this might include data that is wrong without this being obvious, tainting all of your conclusions.